Quiz 1, Modules 1–2 (paper)

Module 1 - Introduction & Math Review · Canvas quiz · 26 Aug 2026

Name _____ Date _____

Question 1 Which of the following is true about a process in which the rate depends on the concentration? Select all that apply.

A. The half-life depends on the concentration of the drug present.
B. The half-life is independent of concentration.
C. The rate equation is $dC/dt = -kC$.
D. The rate equation is $dC/dt = -k$.

Question 2 Which of the following is true about a zero-order process?

A. The elimination rate increases as a function of time.
B. The elimination rate is at its highest when the amount of drug in the body is at its lowest.
C. The elimination rate is independent of the amount of drug in the body.
D. The elimination rate is at its highest when the amount of drug in the body is at its highest.

3a A 165-lb patient received a single IV bolus dose of an antibacterial agent at a level of 30 mg/kg. Blood samples were collected at various time intervals. The elimination of drug seems to follow a first-order model best described by the equation $C = C_0e^{-kt}$. The plasma concentrations (C_p, mcg/mL) were: 0.5 hr, 112; 2 hr, 90.9; 4 hr, 68.9; 8 hr, 39.6; 16 hr, 13.1. What is the half-life of this agent in this patient?

3b A 165-lb patient received a single IV bolus dose of an antibacterial agent at a level of 30 mg/kg. The elimination of drug follows $C = C_0e^{-kt}$, and the plasma concentrations (C_p, mcg/mL) were: 0.5 hr, 112; 2 hr, 90.9; 4 hr, 68.9; 8 hr, 39.6; 16 hr, 13.1. What is the volume of distribution of this agent in this patient?

3c A 165-lb patient received a single IV bolus dose of an antibacterial agent at a level of 30 mg/kg. The elimination of drug follows $C = C_0e^{-kt}$, and the plasma concentrations (C_p, mcg/mL) were: 0.5 hr, 112; 2 hr, 90.9; 4 hr, 68.9; 8 hr, 39.6; 16 hr, 13.1. How much time following administration of the dose is required for 75% of the dose to be eliminated?

Solutions

Her key: the answer and working on her own solved sheet or annotated slide. Worked by the drill: the drill's own working, where her sheet gives no worked value.

Question 1 Her key

Answer: B. The half-life is independent of concentration.; C. The rate equation is dC/dt = −kC.

A rate that depends on concentration is first order, and the first-order half-life is $t½ = \frac{0.693}{k}$: neither term contains a concentration, so the time to halve is the same from any starting level. In her words, "our half-life is 0.693 over K, doesn't change".

dC/dt is the rate, and here it equals k times C, so the rate changes whenever the concentration changes: the definition of first order. As she put it, "DCDT is our rate, and this one says that the rate is equal to our K times concentration".

Question 2 Her key

Answer: C. The elimination rate is independent of the amount of drug in the body.

Zero order means $\frac{dC}{dt} = -k$ with no amount or concentration term, so the same quantity leaves in each unit of time however much drug is present. This is the flipped form of her first-order item, whose key was the rate being highest when the amount is highest.

3a Her key

Answer: 5.006 hr

WHAT THE STEM GIVES

weight	165 lb	*not needed for the half-life*
D_0	30 mg/kg	*not needed for the half-life*
C_p at 0.5 hr	112 mcg/mL	the earlier point of the pair
C_p at 16 hr	13.1 mcg/mL	the later point of the pair
other points	90.9, 68.9, 39.6	*not needed: any pair gives about the same k*

THE WORKING, ONE LINE AT A TIME

SETUP

$$k = \frac{\ln(Cp1 \div Cp2)}{t_2 - t_1}, \text{ then } t½ = \frac{0.693}{k}$$

A first-order decline is a straight line of ln C against time with slope −k, so two points give k, and the half-life follows from k. The first and last points span the most time, which keeps rounding in the printed values from moving k.

ALGEBRA

$$k = \frac{\ln(112 \div 13.1)}{(16 - 0.5)\ \text{hr}} = \frac{\ln 8.5496}{15.5\ \text{hr}} = \frac{2.1459}{15.5\ \text{hr}} = 0.13844\ \text{hr}^{-1}$$

The concentration units cancel inside the ratio, so the logarithm is a pure number; dividing by hours leaves reciprocal hours, the unit of a first-order rate constant.

ALGEBRA

$$t½ = \frac{0.693}{0.13844\ \text{hr}^{-1}} = 5.006\ \text{hr}$$

A pure number over reciprocal hours gives hours. Any other pair of points in the table gives 4.98 to 5.01 hr.

ROUND

$t½ = 5.0$ hr

The stem gives no rounding rule. She accepted 4.97 and said a value near 4.19 means "something's funky".

> **DOES THE NUMBER MAKE SENSE?**
> From 2 to 8 hr the level falls 90.9 → 39.6 mcg/mL, a little more than half in 6 hr, so the half-life is a little under 6 hr; from 8 to 16 hr it falls 39.6 → 13.1, about a third in 8 hr. 5.0 hr fits both.

3b Her key

Answer: 18.75 L

WHAT THE STEM GIVES

weight	165 lb	converted to 75 kg to scale the dose
D_0	30 mg/kg	2250 mg, the numerator
C_p at 0.5 hr	112 mcg/mL	run back to time zero to give C_0
k	0.13844 hr^{-1}	from part a, for the back-extrapolation

THE WORKING, ONE LINE AT A TIME

UNIT

$$\text{weight} = \frac{165\ \text{lb}}{2.2\ \text{lb/kg}} = 75\ \text{kg};\ D_0 = (30\ \text{mg/kg})(75\ \text{kg}) = 2250\ \text{mg}$$

The dose is per kilogram, so the weight in kilograms comes first; the kilograms cancel and leave milligrams.

SETUP

$$C_0 = C_p \cdot e^{+kt} = (112\ \text{mg/L})\ e^{(0.13844)(0.5)} = (112)(1.0717) = 120.03\ \text{mg/L}$$

The first sample was drawn at 0.5 hr, not at time zero, so it is run back half an hour along the same line. She said: "You can't use D_0 and the concentration 30 minutes after the dose was given. So you gotta go back up and find C_0."

ALGEBRA

$$V_D = \frac{D_0}{C_0} = \frac{2250\ \text{mg}}{120.03\ \text{mg/L}} = 18.75\ \text{L}$$

An amount over a concentration leaves a volume: the milligrams cancel and the litre inverts.

ROUND

$V_D \approx 18.8$ L

Her answer was "18.8-ish". A C_0 read from a fitted line through all five points (119.9 mg/L) gives 18.76 L, the same to one decimal place.

> **DOES THE NUMBER MAKE SENSE?**
> 2250 mg over 120.03 mg/L is just under 19 L, a quarter of the 75 kg body weight.

3c Her key

Answer: 10 hr

WHAT THE STEM GIVES

t½	5.0 hr	from part a; two of them
fraction eliminated	75%	25% remaining, two half-lives
D_0	30 mg/kg	*not needed once t½ is known*

THE WORKING, ONE LINE AT A TIME

SETUP

75% eliminated = 25% remaining = ½ × ½: two half-lives

Each half-life leaves half of what was there: 100% → 50% → 25%. Two halvings leave a quarter, which is 75% gone.

ALGEBRA

$t = 2 \times 5.0 \text{ hr} = 10 \text{ hr}$

Two half-lives of 5.0 hr. In her words: "75%, 2 half-lives. 2 times 5 gives us 10."

ALGEBRA

$$\text{check: } t = \frac{\ln(C_0 \div C)}{k} = \frac{\ln 4}{0.13844 \text{ hr}^{-1}} = \frac{1.3863}{0.13844} = 10.01 \text{ hr}$$

C_0 over C is 100 over 25, which is 4, so the natural-log form gives the same time.

> **DOES THE NUMBER MAKE SENSE?**
> 75% gone is two half-lives, and 87.5% would be three; 10 hr is two of 5 hr.
